

agentic-ai-azure-week09-hosting-scale

Week 9 - Hosting, Deployment & Scale

Agentic AI on Azure - Enterprise Master Class | Handout (slides + notes)

Notes

Welcome. This deck covers 'agentic-ai-azure-week09-hosting-scale' - Week 9 - Hosting, Deployment & Scale. Frame the problem and why it matters, then walk the class through the learning goal, the enterprise use case, the architecture/flow, the key concepts, a live run in VS Code, and the architect's takeaways. The repo runs locally with no Azure, so demo it live.

Slide 2 - Learning goal

- Package agents for production scale
- Run durable long-running tasks
- Choose the right Azure compute substrate

Notes

[Learning goal] Walk through these talking points:

- Package agents for production scale
- Run durable long-running tasks
- Choose the right Azure compute substrate

Ask the class: which of these ideas is new to you?

Slide 3 - Enterprise use case

- Black-Friday Customer Service Swarm
- Scales 10 -> 10,000 sessions, scale-to-zero overnight
- Durable refund tasks without blocking requests

Notes

[Enterprise use case] Walk through these talking points:

- Black-Friday Customer Service Swarm
- Scales 10 -> 10,000 sessions, scale-to-zero overnight
- Durable refund tasks without blocking requests

Tip: relate this to a regulated scenario your students know.

Slide 4 - Architecture / flow

- Fast path: FAQ answers inline
- Durable path: refund -> background task
- Status polling at /api/v1/tasks/{id}
- Docker + Gunicorn/Uvicorn for Container Apps

Notes

[Architecture / flow] Walk through these talking points:

- Fast path: FAQ answers inline
- Durable path: refund -> background task
- Status polling at /api/v1/tasks/{id}
- Docker + Gunicorn/Uvicorn for Container Apps

Demo: trace a single request through these steps live.

Slide 5 - Key concepts

- BackgroundTasks now; Durable Functions in prod
- Scale knobs feed KEDA autoscaling
- Decision matrix: Container Apps vs AKS vs Functions

Notes

[Key concepts] Walk through these talking points:

- BackgroundTasks now; Durable Functions in prod
- Scale knobs feed KEDA autoscaling
- Decision matrix: Container Apps vs AKS vs Functions

Open the named file in the repo while you explain each point.

Slide 6 - Run it (VS Code - no Azure needed)

- git clone the repo, then open it in VS Code
- `python -m venv .venv -> activate it`
- `pip install -r requirements.txt`
- `uvicorn app.main:app --reload`
- Open /docs and call POST /api/v1/support
- Runs in MOCK mode - no Azure/keys needed

Notes

[Run it (VS Code - no Azure needed)] Walk through these talking points:

- git clone the repo, then open it in VS Code
- `python -m venv .venv -> activate it`
- `pip install -r requirements.txt`
- `uvicorn app.main:app --reload`
- Open /docs and call POST /api/v1/support
- Runs in MOCK mode - no Azure/keys needed

Pause for questions before moving on.

Slide 7 - Architect's takeaways

- Container Apps is the course default (managed, scale-to-zero)
- Async + worker model + connection pooling
- Cost: token budgets, model routing, caching

Notes

[Architect's takeaways] Walk through these talking points:

- Container Apps is the course default (managed, scale-to-zero)
- Async + worker model + connection pooling
- Cost: token budgets, model routing, caching

Discussion: when would you NOT choose this approach?